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Chemical Bonding and Hybridisation —

Issue #32 specimen: Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

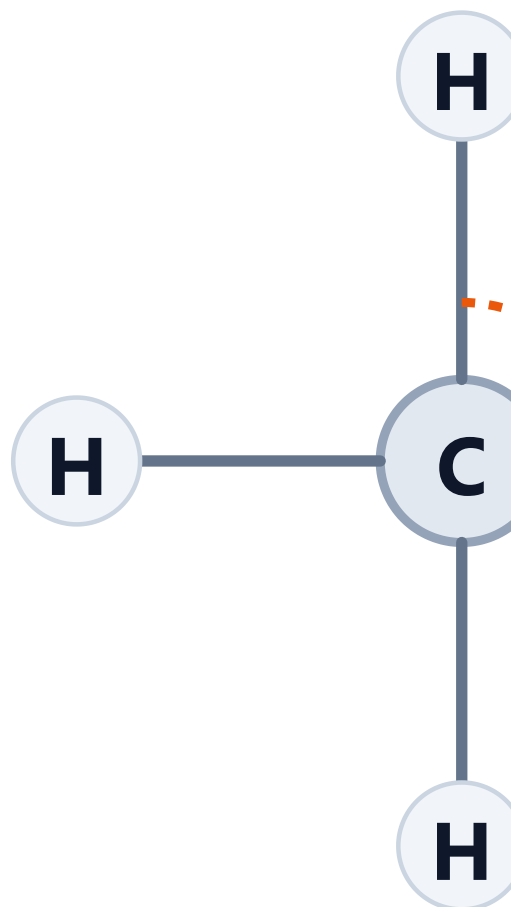
ATTEMPT FIRST

In the introductory valence-bond model, name the hybrid-orbital set used for four equivalent electron-domain directions around a central atom, and state how many hybrid orbitals it contains.

Representation

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Flat 2D cross

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route • open if you want a warning

—

Common wrong route

The superscript is the total number of hybrid orbitals.

Need a hint? • guided support

—

HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Distinguish the hybrid-set name from the number of orbitals in that set.

H1 Representation

Guided

A change of orbital basis preserves the number of independent orbitals.

H2 First move

Guided

Write one valence s plus n valence p orbitals; determine n and the resulting set size yourself.

Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Answer and working

Hide

Reasoning route

1 Understand

Read the requested four equivalent domain directions.

Uses

In the introductory valence-bond model, name the hybrid-orbital set used for four equivalent electron-domain directions around a central atom, and state how many hybrid orbitals it contains.

Why valid: The question supplies a tetrahedral four-direction introductory model.

Result: Four model directions must be represented.

2 Connect

Key move

Select one valence s and three valence p inputs.

Uses Four model directions must be represented.

Why valid: For a second-period centre, these four independent basis functions can be recombined without changing the basis dimension.

Result: $2s + 2p_x + 2p_y + 2p_z$: four inputs.

3 Calculate

Name the recombined set and count its members.

Uses $2s + 2p_x + 2p_y + 2p_z$: four inputs.

Why valid: The name sp^3 records one s and three p inputs; the total number is $1+3$.

Result: sp^3 ; four hybrid orbitals.

4 Interpret

Compare the count with an sp allocation.

Uses sp^3 ; four hybrid orbitals.

Why valid: One s plus one p has two outputs, so the superscript alone cannot be the set size.

Result: sp has two outputs; the four-member sp^3 inventory is consistent.

For second-period atoms, one $2s$ plus three $2p$ orbitals gives four sp^3 hybrids. The superscript counts p inputs, not total outputs.

Independent check

Changed check: an sp set has two hybrids because one s plus one p supplies two basis functions; use this conservation rule on the requested set.

Test the rule on a changed case

An sp² set is proposed to contain two hybrids. How many input and output orbitals should it have? Then apply your method to an sp set.

Your prediction and reason

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

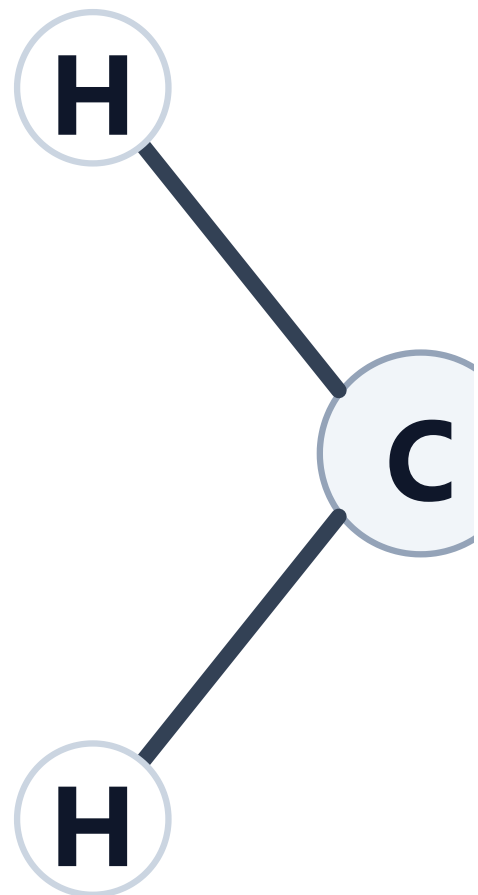
ATTEMPT FIRST

The Lewis structure H₂C=O has a double bond from carbon to oxygen and one single bond from carbon to each hydrogen. Explain why the C=O double bond counts as one electron domain around carbon.

Representation

Stage 1: Lew

Four shared elect



Student trap: con
as 4 separate

Lewis structure

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Common wrong route

Each line of a double bond is a separate local direction.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept Guided

Separate a bonded-atom direction from the number of shared pairs.

H1 Representation Guided

The sigma component and pi component of a multiple bond connect the same atom pair.

H2 First move Guided

Circle the distinct neighbours of carbon, then keep an independent bond-component inventory.

Need the concept again?

Steric electron-domain counting and multiple bonds

Answer and working

Hide

Reasoning route

1 Understand

Make separate neighbour and shared-pair inventories for HCHO.

Uses

The Lewis structure $\text{H}_2\text{C}=\text{O}$ has a double bond from carbon to oxygen and one single bond from carbon to each hydrogen. Explain why the $\text{C}=\text{O}$ double bond counts as one electron domain around carbon.

Why valid: Two H atoms and one O are attached to C; a $\text{C}=\text{O}$ double bond contains two shared pairs.

Result: Three neighbours; two pairs in $\text{C}=\text{O}$.

2 Connect

Key move

Count C-to-O as one local VSEPR domain.

Uses Three neighbours; two pairs in $\text{C}=\text{O}$.

Why valid: A multiple bond connects one pair of nuclei and supplies one bonded-neighbour direction in local domain counting. Sigma and pi density have different spatial symmetry.

Result: One $\text{C}=\text{O}$ domain; this does not place pi density on the internuclear axis.

3 Calculate

Add the two C-H directions.

Uses One $\text{C}=\text{O}$ domain; this does not place pi density on the internuclear axis.

Why valid: Each of the two distinct C-H connections adds one domain and C has no lone pair.

Result: $1+1+1 =$ three C-centred domains.

4 Interpret

Recount carbon directions in HCN.

Uses $1+1+1 =$ three C-centred domains.

Why valid: H-C and C-N connect carbon to two neighbours although C-N has three shared pairs.

Result: Two domains in HCN; shared-pair count and neighbour count are different.

HCHO carbon has three bonded-atom directions. C=O contributes one local domain, although its double bond contains one sigma and one pi component. Pi density is not confined to the internuclear axis.

Independent check

Changed check: a triple bond to N in HCN still supplies one C-to-N direction; carbon also has the H direction.

Test the rule on a changed case

At carbon in HCN, count bonded-atom directions and C-N shared pairs separately. Repeat the two counts for HCHO.

Your prediction and reason

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Direct application

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

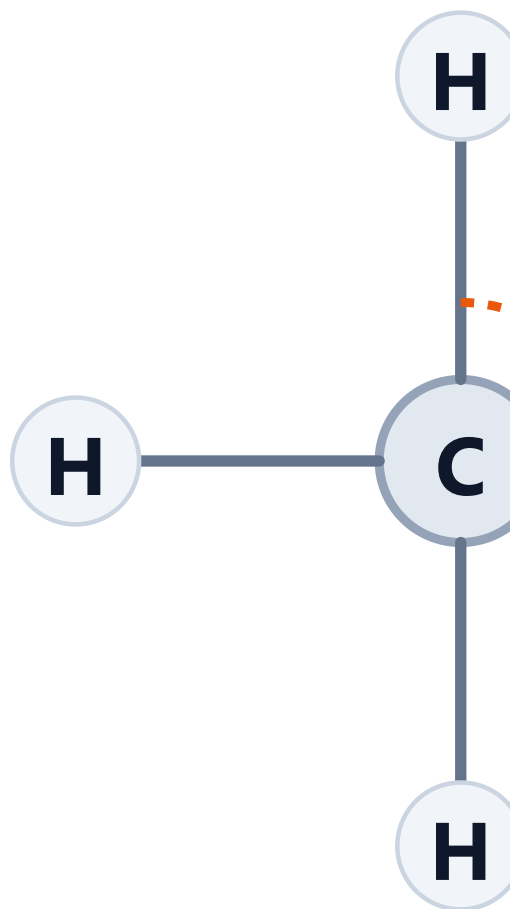
ATTEMPT FIRST

For CH₄, use the known rule for four bonding domains and no central lone pair to assign the carbon hybridisation and electron-domain geometry. Show the domain count.

Representation

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Flat 2D cross

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Common wrong route

The page layout is the molecular geometry.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

The task asks for a local count before the introductory geometry label.

H1 Representation

Guided

Four bonding directions and no central lone pair use the demonstrated four-domain rule.

H2 First move

Guided

Inventory only carbon-centred domains; attach the known rule without reading shape from page angles.

Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Answer and working

Hide

Reasoning route

1

Understand

Inventory CH₄ at carbon.

Uses

For CH₄, use the known rule for four bonding domains and no central lone pair to assign the carbon hybridisation and electron-domain geometry. Show the domain count.

Why valid: The supplied Lewis connectivity has four C-H bonds and no C lone pair.

Result: Four bonding domains; zero central lone pairs.

2

Connect

Key move

Apply the explicitly supplied four-domain rule.

Uses Four bonding domains; zero central lone pairs.

Why valid: The task asks for the known introductory correspondence, rather than an experimental determination of orbital coefficients.

Result: Tetrahedral domain geometry; introductory sp³ description.

3

Calculate

Check the orbital basis ledger.

Uses Tetrahedral domain geometry; introductory sp³ description.

Why valid: One 2s plus three 2p inputs gives four hybrids, matching four model sigma directions.

Result: Four inputs and four outputs; no residual p in this allocation.

4 Interpret

Change one substituent without adding a neighbour.

Uses Four inputs and four outputs; no residual p in this allocation.

Why valid: CH₃Cl still has four carbon-neighbour connections; different bond properties do not constitute extra domains.

Result: CH₃Cl also has four C-centred domains; no exact bond-property claim follows.

CH₄ has four bonding domains, no carbon lone pair, and a tetrahedral introductory sp³ description. A Lewis layout encodes connectivity.

Independent check

Changed check: CH₃Cl still has four bonded neighbours around carbon even if the bonds have different properties.

Test the rule on a changed case

Count domains at C in CH₃Cl and C in CH₄. Does redrawing either as a cross change its count?

Your prediction and reason

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

NH₃ has three N–H single bonds and one lone pair on nitrogen. Translate that information into a labelled diagram showing all four electron domains; distinguish the lone pair from the three bond directions. Do not treat the planar Lewis sketch as the molecular shape.

Representation

NH₃: three N-H bond

Inventory

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Common wrong route

Adding depth marks permits an extra bond or domain.

Need a hint? · guided support

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HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept Guided

Translate three bonds and one lone pair without adding a fifth direction.

H1 Representation Guided

Wedges and dashes encode depth; electron-domain geometry includes lone pairs, molecular shape uses nuclei.

H2 First move Guided

Make four labelled slots first; assign three N-H bonds and one lone pair exactly once.

Need the concept again?

3D Stereochemical representation and qualitative models

Answer and working

Hide

Reasoning route

1 Understand

Count exactly three N-H connections and one N lone pair.

Uses

NH₃ has three N-H single bonds and one lone pair on nitrogen. Translate that information into a labelled diagram showing all four electron domains; distinguish the lone pair from the three bond directions. Do not treat the planar Lewis sketch as the molecular shape.

Why valid: The question supplies the inventory at nitrogen.

Result: Four domains: three bonds and one lone pair.

2 Connect

Key move

Project the four domain directions with a declared depth convention.

Uses Four domains: three bonds and one lone pair.

Why valid: Two ordinary lines may denote in-plane directions; a solid wedge denotes toward and a hashed wedge away. The lone-pair slot must replace one bond slot.

Result: N at centre; exactly three labelled H endpoints and one LP direction.

3 Calculate

Distinguish domain arrangement from nuclear shape.

Uses N at centre; exactly three labelled H endpoints and one LP direction.

Why valid: Domain geometry counts all four slots; molecular shape counts the three bonded H nuclei.

Result: Tetrahedral domain model; trigonal-pyramidal molecular shape, without an exact-angle inference.

4 Interpret

Compare with the four-bond NH_4^+ inventory.

Uses

Tetrahedral domain model; trigonal-pyramidal molecular shape, without an exact-angle inference.

Why valid: Replacing the LP slot with a fourth N-H bond adds a bonded nucleus but no fifth domain.

Result: NH_3 : three H + LP; NH_4^+ : four H. The NH_3 drawing must not contain a fourth H.

NH_3 has N at the centre, exactly three N-H directions and one distinguishable lone-pair domain. Tetrahedral domain arrangement and trigonal-pyramidal molecular shape describe different inventories.

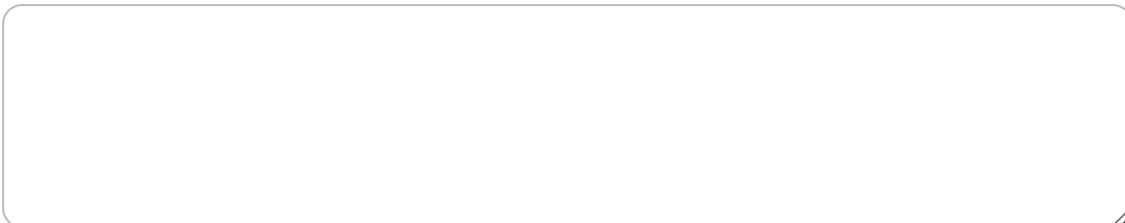
Independent check

Count all four slots in the sketch; replacing the lone-pair slot by an N-H bond changes the inventory to an NH_4^+ teaching comparison, not NH_3 .

Test the rule on a changed case

Draw NH_3 and a tetrahedral four-bond NH_4^+ comparison. Mark which slot differs and recount the H atoms in each.

Your prediction and reason



NH₃: three N-H bond

Use supplied givens only.

Stage 1 of 3

Construct exactly three N-H domains and one distinguishable lone-pair domain.

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

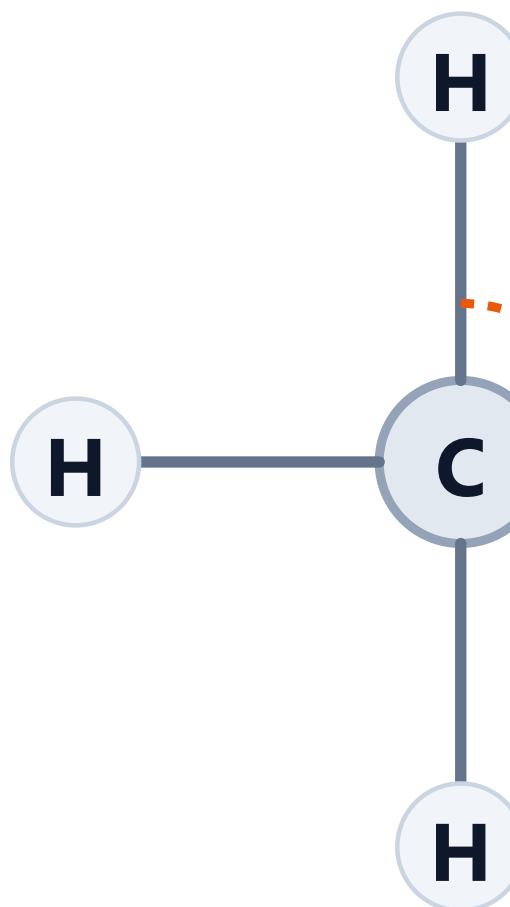
ATTEMPT FIRST

A model is needed only to describe the overall three-dimensional arrangement of electron domains in NH_3 ; it need not describe the detailed orbital overlap. Choose between VSEPR and a hybrid-orbital overlap diagram, and identify what your chosen model predicts and what it leaves unresolved.

Representation

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Flat 2D cross

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal

Common wrong route · open if you want a warning

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coordination geometry
unless specified.

Common wrong route

A familiar chapter keyword selects the model.

Need a hint? · guided support

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Choose a model by the output requested, rather than the chapter title.

H1 Representation

Guided

A model has a defined prediction scope and unresolved questions.

H2 First move

Guided

Compare each candidate against overall domain arrangement versus detailed orbital-overlap information; leave the choice to your answer.

H3 Equation / formal model

Guided

Test your proposed route on this changed case before finalising: Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.

H4 Assembly checkpoint

Guided

Reserve the last judgement for yourself. Try this boundary/transfer task: Choose the needed model when the new task asks for pi-overlap orientation in ethene.

Need the concept again?

3D Stereochemical representation and qualitative models

Answer and working

Hide

Reasoning route

1 Understand

Specify the required output before choosing a model.

Uses

A model is needed only to describe the overall three-dimensional arrangement of electron domains in NH_3 ; it need not describe the detailed orbital overlap. Choose between VSEPR and a hybrid-orbital overlap diagram, and identify what your chosen model predicts and what it leaves unresolved.

Why valid: The task asks for qualitative domain arrangement and expressly excludes detailed orbital overlap.

Result: Required output: qualitative three-dimensional domain geometry.

2 Connect

Key move

Choose VSEPR for that output.

Uses Required output: qualitative three-dimensional domain geometry.

Why valid: VSEPR uses the three bonding domains and central lone pair to describe their approximate spatial arrangement.

Result: Four-domain, approximately tetrahedral arrangement; pyramidal bonded-nucleus shape.

3**Calculate**

Declare unresolved quantities.

Uses

Four-domain, approximately tetrahedral arrangement; pyramidal bonded-nucleus shape.

Why valid: A qualitative domain model does not supply orbital wavefunctions, overlap amplitudes or exact measured angles.

Result: No detailed orbital composition, exact angle or spectrum is determined.

4**Interpret**

Change the required output to axial versus lateral overlap.

Uses No detailed orbital composition, exact angle or spectrum is determined.

Why valid: That question needs orbital orientation/symmetry information beyond a geometry-only model.

Result: An orbital-overlap account is now required; familiarity alone cannot select the model.

VSEPR fits qualitative electron-domain arrangement. An orbital-overlap model addresses orbital participation. VSEPR does not itself calculate an exact experimental angle or full electronic structure.

Independent check

Changed task: detailed axial versus side-on overlap now requires an overlap description; a geometry-only account is insufficient.

Test the rule on a changed case

Task A asks for atom/domain arrangement; task B asks which orbitals overlap along the bond axis. State what information each model must supply.

Your prediction and reason

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For a qualitative orbital sketch, overlap A is end-on along the internuclear axis and overlap B is side-on on opposite sides of that axis. Classify each as sigma or pi overlap and identify the defining feature you used.

Representation

Stage 1: Internuclear Axis Reference

The internuclear axis (z-axis) connects the r

Internuclear axis

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Common wrong route

Turning a drawing on the page changes sigma into pi.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Use the stated internuclear axis to classify the overlap.

H1 Representation

Guided

End-on and side-on are orientation/symmetry descriptions, not molecule-name clues.

H2 First move

Guided

Locate the axis and the density above/below it; name the two classes yourself.

Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Answer and working

Hide

Reasoning route

1

Understand

Mark the line joining the nuclei in each sketch.

Uses

For a qualitative orbital sketch, overlap A is end-on along the internuclear axis and overlap B is side-on on opposite sides of that axis. Classify each as sigma or pi overlap and identify the defining feature you used.

Why valid: Sigma/pi labels refer to overlap relative to this internuclear axis, not to page orientation.

Result: A common geometric reference is declared.

2

Connect

Key move

Classify A from its stated end-on overlap.

Uses A common geometric reference is declared.

Why valid: The supplied overlap is axial; in the introductory sketch the sigma component is symmetric about the bond axis.

Result: A is sigma overlap.

3

Calculate

Classify B from side-on p overlap.

Uses A is sigma overlap.

Why valid: The supplied parallel lateral lobes have a nodal plane containing the axis.

Result: B is pi overlap.

4

Interpret

Rotate the whole picture, including its axis.

Uses B is pi overlap.

Why valid: A rigid rotation preserves all relative orientation and symmetry relations.

Result: Neither classification changes under whole-picture rotation.

Sigma overlap is end-on/axial in the supplied sketch; pi overlap is lateral with a nodal plane containing the axis.

Independent check

Changed check: rotating the entire sketch on the page leaves its classification unchanged relative to the rotated axis.

Test the rule on a changed case

Rotate the whole end-on sketch by 90 degrees on paper. Does it become pi? Compare that with rotating one p orbital relative to its neighbour.

Your prediction and reason

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: 3D Stereochemical representation and qualitative models

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

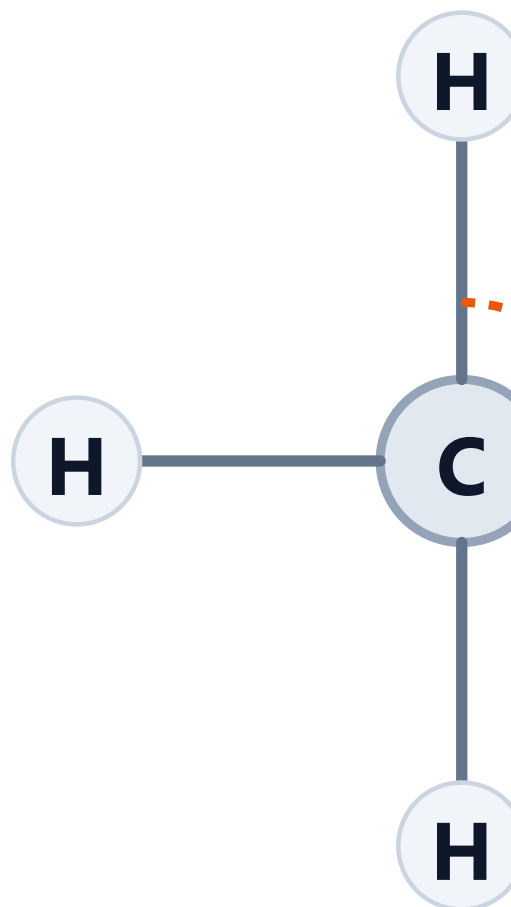
ATTEMPT FIRST

For CH₄, sketch a three-dimensional arrangement of the four C–H bond directions using a wedge/dash convention or another clearly defined convention. Explain how your drawing differs from a flat cross-shaped Lewis sketch.

Representation

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Flat 2D cross

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route • open if you want a warning

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Common wrong route

A Lewis cross proves a square-planar molecule.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Identify whether the error is in connectivity notation or in its interpretation as physical shape.

H1 Representation

Guided

A flat Lewis drawing does not assert that all nuclei are coplanar.

H2 First move

Guided

Keep four C-H connections while assigning explicit towards/away/in-plane conventions.

Need the concept again?

3D Stereochemical representation and qualitative models

Answer and working

Hide

Reasoning route

1 Understand

Retain the four C-H connections from Lewis notation.

Uses

For CH₄, sketch a three-dimensional arrangement of the four C-H bond directions using a wedge/dash convention or another clearly defined convention. Explain how your drawing differs from a flat cross-shaped Lewis sketch.

Why valid: A Lewis cross records connectivity; its page angles do not measure molecular angles.

Result: C at centre with four H endpoints; no bond is added or removed.

2 Connect

Key move

Declare and use a wedge/dash convention.

Uses C at centre with four H endpoints; no bond is added or removed.

Why valid: A solid wedge points toward the viewer, a hashed wedge away; ordinary lines lie in the chosen page plane.

Result: Two in-plane bonds, one toward bond and one away bond represent a tetrahedral model.

3 Calculate

Explain the physical and projected angles separately.

Uses

Two in-plane bonds, one toward bond and one away bond represent a tetrahedral model.

Why valid: The ideal tetrahedral model has about 109.5-degree physical angles, while a projection distorts page angles.

Result: The flat cross does not assert square-planar CH₄; depth belongs to the 3D interpretation.

4 Interpret

Change viewpoint and recount endpoints.

Uses

The flat cross does not assert square-planar CH₄; depth belongs to the 3D interpretation.

Why valid: A rotation or a different projection preserves connectivity and noncoplanar geometry.

Result: Four C-H bonds remain; the drawing is a projection of the same tetrahedron.

For CH₄ a wedge/dash projection represents four tetrahedral bond directions. Projected page angles are not all physical 109.5-degree angles.

Independent check

Changed check: rotate the paper or choose another viewing direction; the atom connectivity and noncoplanar model must stay unchanged.

Test the rule on a changed case

Does a cross-shaped Lewis CH₄ drawing assert square-planar CH₄? Does adding wedge/dash notation change the number of bonds?

Your prediction and reason

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

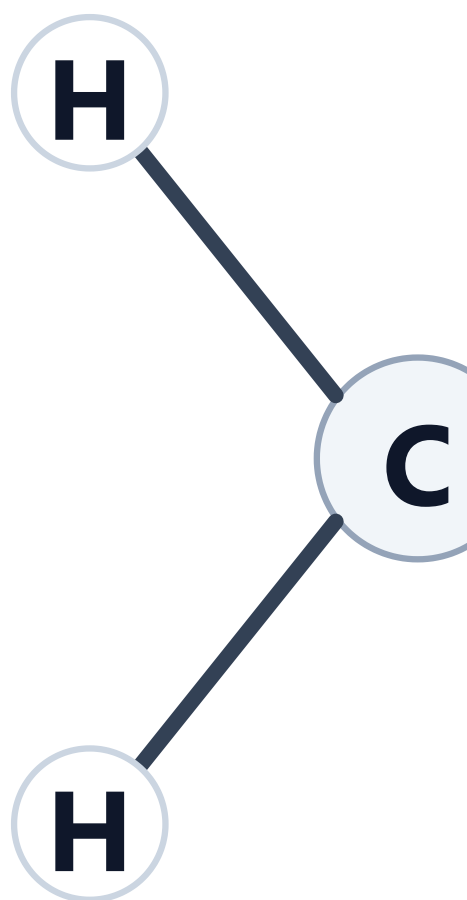
A nitrogen atom in NF₃ has three N–F single bonds and one lone pair. Each fluorine

Representation

also has lone pairs. Count the electron domains around nitrogen and state which lone pairs belong in that central-atom count.

Stage 1: Lew

Four shared elect



Student trap: con
as 4 separate

Lewis structure

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

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Common wrong route

Terminal-atom lone pairs add domains at the central atom.

Need a hint? · guided support

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HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

The counting centre is nitrogen, not every atom in NF_3 .

H1 Representation

Guided

Local domains consist of bonds and lone pairs at the selected atom.

H2 First move

Guided

Circle nitrogen before counting; keep terminal-fluorine pairs in a separate inventory.

Need the concept again?

Steric electron-domain
counting and multiple bonds

Answer and working

Hide

Reasoning route

1 Understand

Choose nitrogen as the counting centre.

Uses

A nitrogen atom in NF₃ has three N–F single bonds and one lone pair. Each fluorine also has lone pairs. Count the electron domains around nitrogen and state which lone pairs belong in that central-atom count.

Why valid: The requested domains are centred on N, not on the entire NF₃ molecule.

Result: Three N–F bonding directions are included.

2 Connect

Key move

Assign each lone pair to its own atom.

Uses Three N–F bonding directions are included.

Why valid: N has one lone pair; terminal F lone pairs belong to F-centred inventories.

Result: One N lone pair included; terminal F lone pairs excluded from the N count.

3 Calculate

Add the local entries.

Uses One N lone pair included; terminal F lone pairs excluded from the N count.

Why valid: The N inventory has three bonded directions plus one central lone pair.

Result: $3 + 1 =$ four nitrogen-centred domains.

4 Interpret

Substitute H for F in the local comparison.

Uses $3+1 =$ four nitrogen-centred domains.

Why valid: NH₃ has the same three-bond/one-central-pair N inventory even though terminal atoms differ.

Result: NH₃ also has four N domains; terminal pairs were not required for that total.

NF₃ has three nitrogen-to-fluorine directions and one nitrogen lone pair: four N-centred domains. Terminal F lone pairs do not add N-centred domains.

Independent check

Changed check: substituting H for F leaves the local three-bond-plus-lone-pair N inventory unchanged in the introductory model.

Test the rule on a changed case

Count N-centred domains in NH₃ and NF₃. Then state where the F lone pairs are located.

Your prediction and reason

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Steric electron-domain counting and multiple bonds

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric electron-domain counting and multiple-bond constraints

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

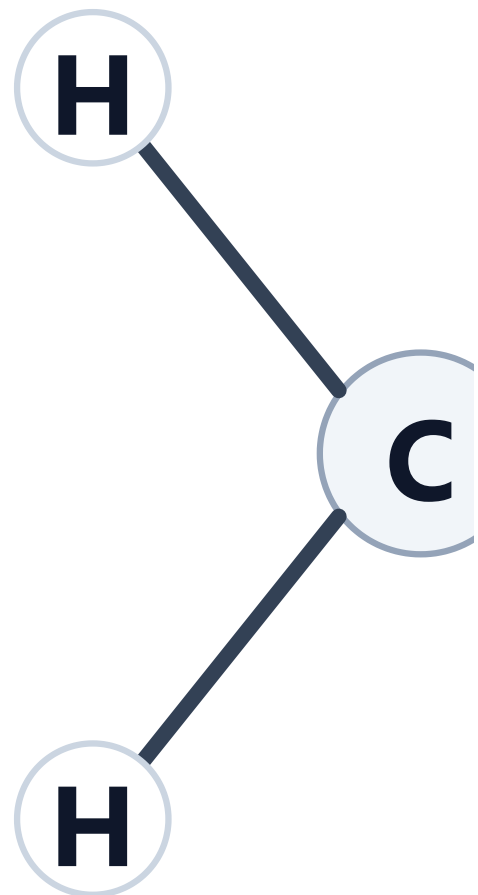
ATTEMPT FIRST

A student counts the double bond in $\text{H}_2\text{C}=\text{O}$ as two separate directions and predicts four domains around carbon. Locate the counting error and give the corrected domain count.

Representation

Stage 1: Lew

Four shared elect



Student trap: con
as 4 separate

Lewis structure

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route · open if you want a warning

—

Common wrong route

Bond multiplicity equals neighbour-direction multiplicity.

Need a hint? · guided support

—

HINT LADDER · REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept Guided

Critique the counting operation, rather than merely giving a corrected label.

H1 Representation Guided

Bond multiplicity and neighbour directions are different ledgers.

H2 First move Guided

Show the faulty ledger beside a neighbour-only ledger; complete the corrected count yourself.

Need the concept again?

Steric electron-domain counting and multiple bonds

Answer and working

Hide

Reasoning route

1 Understand

Identify the faulty C=O line-counting entry.

Uses

A student counts the double bond in $\text{H}_2\text{C}=\text{O}$ as two separate directions and predicts four domains around carbon. Locate the counting error and give the corrected domain count.

Why valid: Two drawn bond lines denote two bond components to one O neighbour.

Result: The student counted one neighbour twice.

2 Connect

Key move

Replace bond-line counting with local neighbour counting.

Uses The student counted one neighbour twice.

Why valid: In VSEPR a multiple bond supplies one bonded-neighbour domain; pi density is lateral rather than an extra atom direction.

Result: C=O contributes one domain, not two.

3 Calculate

Compute the corrected HCHO total.

Uses C=O contributes one domain, not two.

Why valid: There are two distinct H neighbours, one O neighbour and no C lone pair.

Result: $2+1+0 =$ three C-centred domains.

4 Interpret

Apply the operation to CO_2 .

Uses $2+1+0 = \text{three C-centred domains.}$

Why valid: Two C=O bonds connect C to two O neighbours; four drawn bond lines do not imply four local domains.

Result: CO₂ carbon has two domains in the introductory model.

Counting both C=O lines as separate domains confuses bond components with one bonded-atom direction. HCHO carbon has three domains.

Independent check

Changed check: HCN carbon has two neighbour directions even though C-N is triple.

Test the rule on a changed case

A learner counts HCN carbon as four directions from one C-H line plus three C-N lines. Repair the operation and apply it to HCHO.

Your prediction and reason

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Hybridisation, orbital conservation, and overlap symmetry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Hybrid-orbital mapping, orbital symmetry, and 3D stereochemical representation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

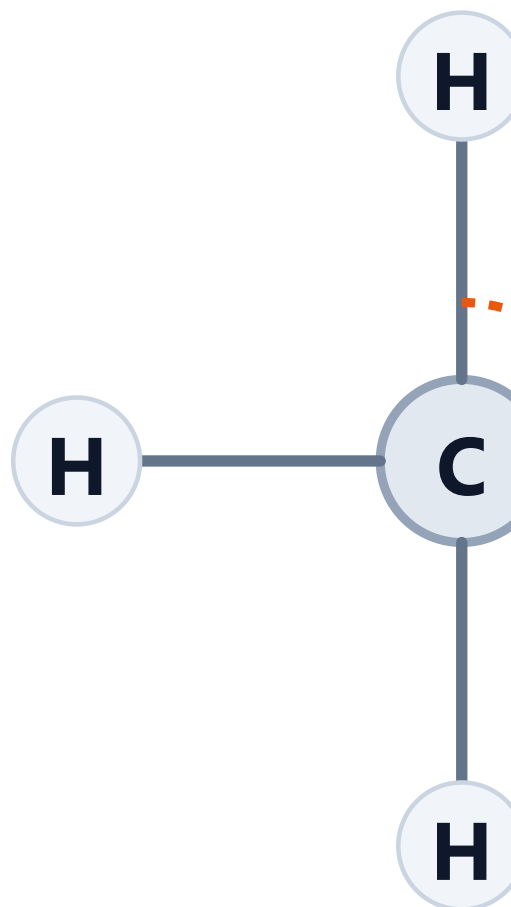
ATTEMPT FIRST

Representation

The benchmark supplies this table: BF₃ has three bonding domains and no central lone pair; CH₄ has four bonding domains and no central lone pair. Match each central atom to the introductory sp² or sp³ model, and check that the number of hybrid-orbital directions agrees with the domain count.

Stage 1: 2D Lewis C

Topological connectivity (F



Misleading: 2D drawing forc

Flat 2D cross

Conditions

- Introductory valence-bond model and VSEPR rules apply; assume ideal coordination geometry unless specified.

Common wrong route • open if you want a warning

—

Common wrong route

sp2 has two hybrids and sp3 has three.

Need a hint? • guided support —

HINT LADDER • REVEAL ONLY WHAT YOU NEED

Guided support

H0 Key concept

Guided

Map each set name to an orbital inventory, not its superscript alone.

H1 Representation

Guided

One s plus n p inputs gives n+1 hybrid outputs.

H2 First move

Guided

Write separate input/output ledgers for BF₃ and CH₄ before matching labels.

Need the concept again?

Hybridisation, orbital conservation, and overlap symmetry

Answer and working

Hide

Reasoning route

1

Understand

Read the two supplied domain inventories.

Uses

The benchmark supplies this table: BF₃ has three bonding domains and no central lone pair; CH₄ has four bonding domains and no central lone pair. Match each central atom to the introductory sp² or sp³ model, and check that the number of hybrid-orbital directions agrees with the domain count.

Why valid: The task supplies three B bonding domains and four C bonding domains, with no central lone pairs.

Result: BF₃: three; CH₄: four.

2 Connect

Key move

Match each inventory to a conserved valence-basis allocation.

Uses BF₃: three; CH₄: four.

Why valid: One s plus two p yields three hybrids; one s plus three p yields four.

Result: BF₃: sp², three hybrids; CH₄: sp³, four hybrids.

3 Calculate

Match the ideal domain arrangement within this introductory model.

Uses BF₃: sp², three hybrids; CH₄: sp³, four hybrids.

Why valid: Three equivalent model directions are planar; four are tetrahedral. These labels are model descriptions.

Result: BF₃: trigonal planar; CH₄: tetrahedral.

4 Interpret

Reconstruct an sp inventory as a third case.

Uses BF₃: trigonal planar; CH₄: tetrahedral.

Why valid: One s plus one p provides two inputs and therefore two hybrid outputs.

Result: The set-size sequence is sp=2, sp²=3, sp³=4, rather than the superscripts 1,2,3.

BF₃: three domains and three sp² hybrids. CH₄: four domains and four sp³ hybrids. For these centres use valence 2s/2p language rather than ambiguous 1s + 2p shorthand.

Independent check

Changed check: sp uses two inputs and two hybrid outputs, providing a third case for the conservation rule.

Test the rule on a changed case

Compare proposed totals $sp=1$, $sp^2=2$, $sp^3=3$ against basis-function conservation. Give the corrected sequence.

Your prediction and reason